The Vol Surface Construction Playbook

From Scattered Quotes to Arbitrage-Free Surfaces

SVI, SSVI, Static Arbitrage, and Dupire Local Volatility Made Reproducible

Amit Kumar Jha

Desk2Quant — Quant Developer & Model Risk Quant

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Every worked example in this note is numerically validated. A companion executable notebook is included with the purchase.

4.2 The SSVI ansatz

Definition: Surface SVI (Gatheral–Jacquier)

$$w(k,\theta) = \frac{\theta}{2}\left[1 + \rho\,\varphi(\theta)\,k + \sqrt{\left(\varphi(\theta)\,k + \rho\right)^2 + (1-\rho^2)}\right],$$

where $\theta(T) = w(0,T)$ is the ATM total variance (fitted freely as a function of T), $\rho \in (-1,1)$ is a global skew parameter, and $\varphi(\theta) > 0$ is a *function of* θ that controls how the smile bends across tenors.

SSVI has a handful of free parameters in practice: ρ, the $T \mapsto \theta$ curve (e.g. fitted at the ATM tenors), and the parameters of $\varphi(\cdot)$. The pay-off is a surface whose slices are all SVI and whose no-arbitrage properties reduce to two inequalities on $\theta\varphi(\theta)$.

4.3 SSVI slices are SVI slices

A subtlety that trips up more than one implementation: SSVI is written in the *natural SVI* parametrisation $w = \Delta + \frac{\omega}{2}\left(1 + \rho\xi k + \sqrt{(\xi k + \rho)^2 + 1 - \rho^2}\right)$ with $\Delta = 0$, $\omega = \theta$, $\xi = \varphi(\theta)$. Converting to the raw SVI parameters used by fitter code requires care with the σ term:

Core Formula: Slice mapping (raw SVI parameters from an SSVI slice)

$$a = \frac{\theta}{2}(1-\rho^2), \qquad b = \frac{\theta\,\varphi}{2}, \qquad \rho^{\text{SVI}} = \rho,$$
$$m = -\frac{\rho}{\varphi}, \qquad \sigma = \frac{\sqrt{1-\rho^2}}{\varphi}.$$

Warning: A common wrong mapping

Several public implementations write $\sigma = 1/\varphi$. Expanding both forms shows the correct term is $\sigma = \sqrt{1-\rho^2}/\varphi$: the raw-SVI square root must reproduce $\sqrt{(\varphi k + \rho)^2 + 1 - \rho^2}$ after scaling, which forces σ to carry the $\sqrt{1-\rho^2}$ factor. Verification: at $\theta = 0.0256$, $\varphi = 4.327$, $\rho = -0.70$ the correct mapping reproduces the SSVI slice to machine precision at every k; the $\sigma = 1/\varphi$ version is off by $\sim$ 11% of variance at the wings. This note's errata list (Appendix C) records the correction explicitly.

Example: Numeric verification of the mapping

$T = 1\text{y}$, $\theta = 0.0256$, $\varphi = \eta\theta^{-\gamma} = 1.2 \cdot (0.0256)^{-0.35} = 4.327$.

k	SSVI $w(k)$	raw-SVI $w(k)$ (correct σ)	difference
-0.30	0.05160	0.05160	$< 10^{-14}$
0.00	0.02560	0.02560	$< 10^{-14}$
$+0.30$	0.01309	0.01309	$< 10^{-14}$

4.4 The power-law parametrisation

Definition: Power-law SSVI

$$\varphi(\theta) = \eta\,\theta^{-\gamma}, \qquad \eta > 0,\ 0 < \gamma < 1 \text{ (practically)}.$$

The power law is the standard practical choice: two parameters, and the no-arbitrage theorems become explicit inequalities that are trivial to check — which is precisely why it dominates production use.

Core Formula: Density proportional to the butterfly functional

$$\frac{\partial^2 C}{\partial K^2}(K,T) = \frac{e^{-rT}}{K\sqrt{2\pi\, w}}\, e^{-d_1^2/2}\, g[w],$$

where

$$g[w](k) = \left(1 - \frac{k\, w_k}{2w}\right)^2 - \frac{w_k^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w_{kk}}{2}.$$

The prefactor $e^{-d_1^2/2}/(K\sqrt{2\pi w})$ is strictly positive, so:

Theorem: Butterfly no-arbitrage (slice)

A slice is free of butterfly arbitrage if and only if $g[w](k) \geq 0$ for all strikes in the slice. Strict positivity gives a strictly positive density.

Economic reading: $g[w]$ is (up to constants) the price of an infinitesimal butterfly, normalised. A negative pocket of $g[w]$ is a butterfly trade with negative cost — free money for whoever quotes it.

5.3 What $g[w]$ sees: a visual

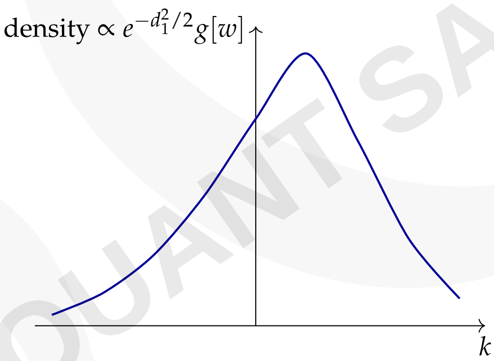

Figure 5.1: Typical risk-neutral density implied by an equity-style smile: skewed left, positive everywhere. Any dip below the axis is a butterfly arbitrage.

5.4 The audit

The practical audit evaluates $g[w]$ on a dense grid of the slice (and, in Module 6, across tenors) using numerical derivatives in k:

```python
import numpy as np

def butterfly_g(w, k):
    """Gatheral-Jacquier butterfly functional on a 1-D k-grid."""
    dk  = k[1] - k[0]
    wk  = np.gradient(w, k)
    wkk = np.gradient(wk, k)
    return (1 - k*wk/(2*w))**2 - (wk**2/4)*(1/w + 0.25) + wkk/2

def audit_surface(W, K):
    """W: (n_T, n_k) total-variance grid. Returns global min of g[w]."""
    gmin = np.inf
    for w_slice in W:
        g = butterfly_g(w_slice, K)
        gmin = min(gmin, g.min())
    return gmin
```

Listing 5.1: Butterfly audit of a surface slice

Core Formula: Coordinate relation for the time derivative

$$(\partial_T w)\big|_K = (\partial_T w)\big|_k - (r-q)\, w_k.$$

Derivation: with $K = F(T)e^k$ fixed, differentiate $k = \ln(K/F(T))$: $0 = \partial_T k|_K + (r-q)$, so $\partial_T|_K = \partial_T|_k - (r-q)\partial_k$. Applied to $w(k,T)$ this gives the formula.

Warning: Mixing conventions is a classic P&L leak

A "small" sign slip here changes local volatility by a drift-sized amount — negligible at $r - q = 0.8\%$ and shallow smiles, but material in high-rate currencies or steep short smiles (Module 8 quantifies the reprice damage). Desks test for this by reprising liquid calls under the local-vol surface (Modules 7–8); a systematic reprice bias after fixing everything else is almost always this term.

6.3 The audit

```python
# W: (n_T, n_k) total variance on a (T,k) grid; T: 1-D tenor grid
dW = np.gradient(W, T, axis=0)        # (dT w) evaluated at FIXED k
assert dW.min() >= -1e-12, "calendar arbitrage present"
```

Listing 6.1: Calendar audit, fixed-k convention

Example: Validated audit (running example)

Same surface as Module 5:

$$\min_{\text{grid}} \partial_T w\big|_k = +0.0123 \geq 0 \quad \Rightarrow \quad \text{PASS.}$$

The companion notebook also runs the negative control: setting $\gamma = 1.4 > 1$ (outside the SSVI calendar domain) produces negative $\partial_T w$ at deep strikes, exactly as Module 4's theorem predicts.

6.4 Why PCA surfaces carry standing arbitrage

Recent work on the geometry of volatility surfaces (arXiv:2608.05198, 2026) proves a striking "boundary incompatibility" result: at an active arbitrage-constraint boundary, a non-degenerate Gaussian (PCA-style) innovation exits the admissible set with probability tending to $\frac{1}{2}$ as the mesh refines. Translation: **a plain PCA factor model of the implied-vol surface generates static arbitrage half the time per step, forever**, unless the generator respects the surface's geometry (projection onto the tangent space, reflection, or arbitrage-aware parameterisations such as SSVI).

Practitioner Insight: Arbitrage checking is risk, not compliance

On a desk, an arbitrage-positive surface is not a footnote — it is a trading signal and a validation failure. Every model staged on top of the surface (local vol, exotic pricing, P&L explain) inherits its defects. The modules ahead assume you audit before you build.

After This Module: After this module

✓ I can state the calendar condition and audit it on a grid. ✓ I can derive and apply the fixed-k vs fixed-K drift correction. ✓ I can explain why PCA surface models need arbitrage-aware generation.

Core Formula: Dupire's forward equation

$$\frac{\partial C}{\partial T} = \frac{1}{2}\,\sigma_L^2(K,T)\,K^2\frac{\partial^2 C}{\partial K^2} - (r-q)\,K\,\frac{\partial C}{\partial K} - q\,C.$$

Solving for the local variance:

$$\sigma_L^2(K,T) = \frac{\dfrac{\partial C}{\partial T} + (r-q)K\dfrac{\partial C}{\partial K} + q\,C}{\dfrac{1}{2}\,K^2\,\dfrac{\partial^2 C}{\partial K^2}}.$$

The formula applies whenever the surface is arbitrage-free (so the denominator, proportional to the density, is positive); it blows up exactly where arbitrage or noise appears.

7.3 The total-variance form

In (k, w) coordinates, with $k = \ln(K/F(T))$, the same object becomes Gatheral's form, which lives on surface derivatives already computed by the audits of Modules 5–6:

Core Formula: Local volatility from total variance

$$\sigma_L^2(k,T) = \frac{\partial_T w|_k}{1 - \dfrac{k}{w}\,w_k + \dfrac{1}{4}\Big(-\dfrac{1}{4} + \dfrac{1}{w}\Big)w_k^2 + \dfrac{1}{2}\,w_{kk}}.$$

The denominator is exactly the butterfly functional $g[w]$ from Module 5:

$$\sigma_L^2 = \frac{(\partial_T w)|_k}{g[w]}.$$

This is the cleanest way to remember both: local vol squared is *time slope over butterfly functional*. The numerator is the calendar condition's object; the denominator is the butterfly condition's object — every arbitrage audit you ran is exactly the machinery local vol needs.

Why the forms agree

The change of variables $C(K,T) \to w(k,T)$ maps option-price derivatives to w-derivatives; in particular $\partial_T|_K = \partial_T|_k - (r-q)\partial_k$ (Module 6's trap), and the call-price second derivative $\frac{1}{2}K^2 C_{KK}$ maps to the $g[w]$ denominator. Both formulas are the same statement in two coordinate systems.

7.4 The convention trap, again

Warning: Which $\partial_T w$ goes in?

The numerator must be the fixed-k derivative $(\partial_T w)|_k$. Code that differentiates the surface in T along *fixed strikes* computes $(\partial_T w)|_K$ and is off by $(r-q)w_k$. With $r - q = 0.8\%$ and the running smile, $w_k \approx -0.003$ at $k = 0.3$, so the error is $\approx 2.6 \times 10^{-5}$ in w — around 1%–2% of the local variance at $T = 1$y, invisible to casual review but persistent across strikes (one of the two classic sources of systematic reprice bias; Module 8's experiments isolate the other).

7.5 Code: local volatility from a validated surface

9.3 Practical playbook

1. Build on a *rolling short-window* T-axis (e.g. last 4–8 hours), not absolute time-of-day, because the relevant horizon is time-to-expiry.
2. Anchor the near end: blend the 0DTE bucket with the shortest listed tenor so the surface does not dangle at $T \approx 0$.
3. Treat the first hours after open as extra-noisy: wider wings, more arbitrage pockets, wider spreads; downweight them.
4. Use SSVI tails for extrapolation (never fit sparse far-wing quotes).
5. Re-run the Module 5–6 audits at high frequency: 0DTE pockets are short-lived and reappear as quotes refresh.

After This Module: After this module

✓ I can list 0DTE's three construction problems and the fixes. ✓ I know the NBBO SPY benchmark and why it is the standard testbed.

9.4 Visualising the compressed 0DTE axis

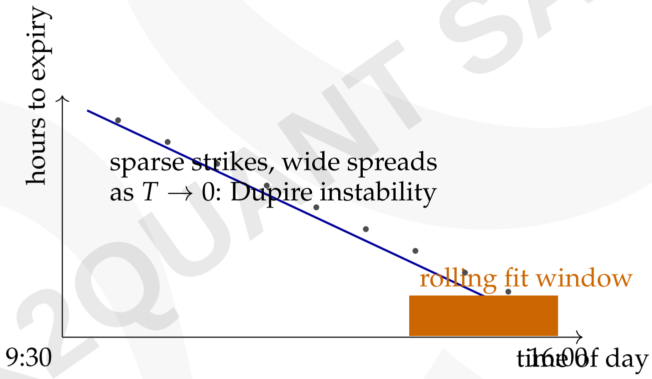

Figure 9.1: The 0DTE construction problem: hours-to-expiry shrinks through the day (blue line), the fitted window rolls (orange band), and the quotes (dots) stay sparse and wide until the bell. Every Module 9 fix targets one of these three facts.

Direct Proportionality: 0DTE noise budget

With h hours to expiry, quote noise and arbitrage-pocket density scale as

$$\text{quote noise} \propto \frac{1}{h}, \qquad \text{pocket density} \propto \frac{\text{bid–ask width}}{(\text{strike spacing})^2},$$

and the local-vol instability grows as

$$\sigma_L \text{ instability} \propto \frac{1}{\sqrt{T}} \qquad (T \to 0).$$

Why it matters: downweighting post-open quotes is literally weighting by h; blending with the short listed tenor is anchoring the $1/\sqrt{T}$ blow-up.

Q5. State the SSVI power-law no-arbitrage domain.

$\gamma \le 1$ for calendar; $\eta(1+|\rho|)\theta^{1-\gamma} \le 4$ for all fitted θ (binding at $\theta_{\max}$ when $\gamma < 1$) for butterfly. $\gamma = 1$ is the flat/Heston-like limit. *Testing: precision with inequalities and the binding-constraint logic.*

Q6. Write Dupire's formula and explain why it is unstable at short expiry / far OTM.

$\sigma_L^2 = [C_T + (r-q)KC_K + qC]/[\frac{1}{2}K^2 C_{KK}]$. At short T and in the wings both numerator and denominator $\to 0$; discretised ratios amplify noise, so T-floors, smooth-family differentiation, and wing caps are required (Module 8). *Testing: can you pair a formula with its numerics?*

Q7. What is the difference between $(\partial_T w)|_K$ and $(\partial_T w)|_k$?

$(\partial_T w)|_K = (\partial_T w)|_k - (r-q)w_k$; the market surface lives in K, the theory in k. Mixing them biases local vol systematically. *Testing: do you know the most common convention bug in production?*

Q8. How would you validate a local-vol surface in production?

Reprice liquid calls under the local-vol process by Monte Carlo and require closure within ~ 2 standard errors; audit butterfly/calendar first; check the drift convention. *Testing: do you have a closure standard, not just a warm feeling?*

Q9. Why do 0DTE surfaces need special treatment?

Sparse strikes, wide spreads, compressed intraday T-axis, Dupire instability at $T \approx 0$; rolling T-axis, blending with the short listed tenor, downweighting post-open quotes. *Testing: do you know the modern market's hardest construction problem?*

Q10. A colleague's surface passes all checks but exotics misprice by 30 bps of vol. What do you check?

(1) Wings beyond the fitted range (garbage fit vs extrapolation), (2) the local-vol conversion convention (fixed-k), (3) denominator floors creating zero-vol pockets, (4) forward-smile behaviour (local-vol flattening) for cliquets/forward-starts, (5) regime/calibration staleness. *Testing: structured diagnostic thinking under pressure.*

Q11. How do you map an SSVI slice to raw SVI parameters?

$a = \frac{\theta}{2}(1-\rho^2)$, $b = \frac{\theta\varphi}{2}$, $\rho^{\text{SVI}} = \rho$, $m = -\rho/\varphi$, and — the detail — $\sigma = \sqrt{1-\rho^2}/\varphi$ (not $1/\varphi$). *Testing: attention to the ρ-dependent normalisation in σ.*

Q12. How are the two Dupire formulas related?

Both compute the same object. The price form uses (C_T, C_K, C_{KK}); the total-variance form uses

$$\sigma_L^2 = \frac{(\partial_T w)|_k}{g[w]}.$$